Kadi-APY Workshop

Automating everyday data management tasks in Kadi4Mat with Python and the command line

DATE AND TIME	FORMAT	MEETING LINK
Friday, 13 November 2026 **14:00-16:00 (CET)**	**Online, hands-on** **in Google Colab**	**[FILL IN]**

About the workshop

kadi-apy is the Python library and command line tool of Kadi4Mat. It allows routine steps of the web interface, such as creating records, uploading files, adding metadata and searching, to be scripted. This is useful whenever the same steps are repeated for many files, for example after a measurement series.

We apply it to a tensile test campaign with three steel specimens (synthetic data): the result files are uploaded, calculated values such as the yield strength are stored as metadata, and the data is found and plotted again. The same steps apply to other instruments and file types. No prior experience with kadi-apy or Python is required.

Schedule

Each part consists of a complete example, a reference of the functions used with links to the kadi-apy documentation, and an exercise.

TIME	BLOCK	CONTENT
14:00-14:10	**Welcome and demonstration**	Kadi4Mat terms used in the workshop (record, metadata, collection, tag). Live demonstration of the final result: three specimens uploaded from a folder with one script.
14:10-14:20	**Part 1: Kadi setup check, kadi-apy documentation**	Connection test. How to find and read functions and commands in the kadi-apy documentation.
14:20-14:35	**Part 2: Command line**	Create a record, upload a file, add metadata and tags with single commands. Suited for quick one-off steps and shell scripts.
14:35-14:55	**Part 3: Python**	The same steps in a Python script, the basis for processing many files at once.
14:55-15:00	Break	
15:00-15:20	**Part 4: Automating a test campaign**	One script for a folder of result files: records with calculated values as metadata, files, a collection and links to the testing machine.
15:20-15:30	**Part 5: Search and plot**	Find records by tag, download their files and plot the stress-strain curves.
15:30-16:00	**Discussion**	Own use cases of the participants, which kadi-apy functions would automate them, open questions.

Preparation and material

Before the workshop (about 20 minutes): create a personal access token in Kadi4Mat (kadi4mat.iam.kit.edu, Settings > Access tokens), open the notebook in Google Colab, store the token as a Colab secret named `KADI_PAT` and run the setup cells. A step-by-step checklist is in the participant guide.

Notebook (Colab) | github.com/msishuvo/thematic-workshop-kadiAPY | kadi-apy.readthedocs.io